



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SHREE RENGGA AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

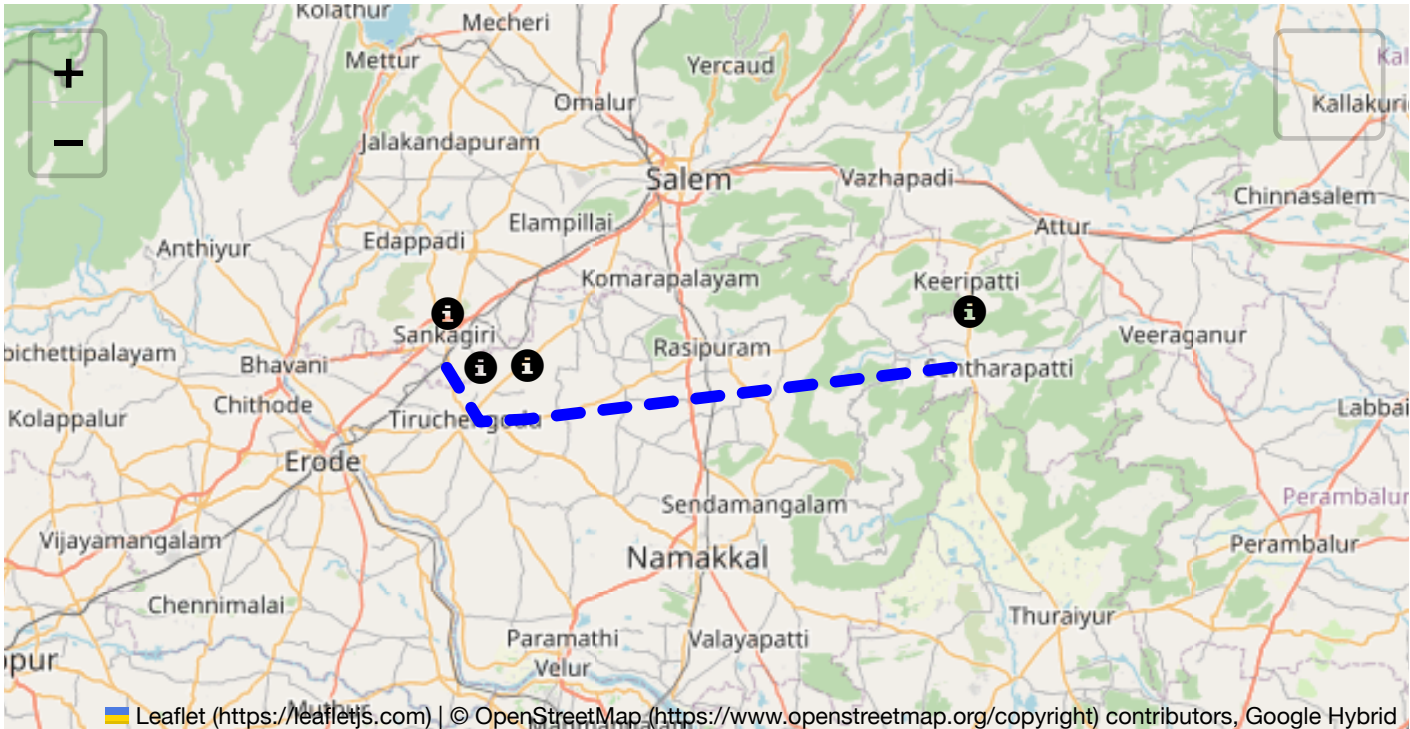
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 86.73 km
Estimated Duration: 2.0 hours
Adjusted Duration (Heavy Vehicle): 2.6 hours

Start: (11.4381, 77.8734)
End: (11.43848, 78.485985)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to Thammampatti via Namakkal and Kavundampalaiyam covers approximately 86.73 kilometers. It utilizes local roads and State Highway 79, which are generally rural but crucial for regional connectivity. This route mainly passes through agricultural areas, small towns, and village clusters, with varying traffic conditions depending on proximity to larger towns like Namakkal.

Weather Conditions

The area experiences a tropical climate with moderate to heavy rainfall from June to November, posing a risk of slippery roads and reduced visibility. Summers are hot and dry, potentially leading to heat-related vehicle issues, while winters are mild. Drivers should prepare for fog during early mornings and late nights from December to February.

Traffic Patterns

Traffic is moderately dense near Namakkal, especially during peak hours (8:00-10:00 AM and 5:00-7:00 PM). Congestion is mainly due to local commuters and goods transportation. The road sections close to the agricultural markets can experience increased traffic due to the movement of goods vehicles.

Road Quality and Infrastructure

Road quality fluctuates, with main highways mostly in good condition. However, some sections might have potholes and uneven surfaces, particularly near village entries and where minor roads intersect. Infrastructure, like signage and lighting, might be insufficient in more remote areas, increasing risks in poor visibility or night travel.

Alternative Routes for Emergencies

For emergencies, consider taking the NH44 route for more reliable road conditions and amenities, though it might add to the travel distance. This highway offers better connectivity to nearby towns and easier access to emergency services.

Local Regulations Affecting Hazardous Material Transport

Tamil Nadu regulations require special permits for transporting hazardous materials, including adherence to specific routes, times of transit, and emergency preparedness. Always check for updates and compliance requirements prior to transit.

Historical Incidents

While specific incident data for this route is limited, the region has reported occasional accidents involving heavy vehicles due to overloading and poor road maintenance. Incidents involving hazardous materials are rare but typically occur when safety protocols are not followed.

Environmental Considerations

The route runs through agricultural zones and near water bodies, necessitating caution to prevent contamination from spills. It's crucial to identify and avoid disturbing ecologically sensitive regions, particularly in monsoons when soil erosion is a concern.

Communication Coverage

Mobile network coverage is generally stable in and around towns and major roads but might be patchy in more rural segments. Drivers should familiarize themselves with dead zones and carry alternative means of communication, like satellite phones, for emergencies.

Estimated Emergency Response Times

Response times can vary significantly:

- Urban areas like Namakkal: approximately 20-30 minutes
- Rural areas or along SH 79: 45-60 minutes Preparation through local emergency contacts is essential to improve reaction time in these areas.

Overall Risk Assessment

Overall, the route presents moderate risk levels under typical conditions, primarily influenced by weather variability, road quality, and traffic density. Key recommendations include proactive weather monitoring, adherence to safer and regulated routes for hazardous material transport, and thorough pre-trip safety checks for vehicles. Enhanced training for drivers on emergency procedures and regular updates on local regulations will further mitigate risks.

Drivers are encouraged to keep updated with real-time traffic and weather information and maintain open communication with their dispatch center to adjust plans as necessary.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44019, 77.87547	15 KM/Hr
3	Turn	Medium	11.39685, 77.88861	30 KM/Hr
4	Turn	Medium	11.38376, 77.89914	30 KM/Hr
5	Turn	Medium	11.38342, 77.89975	30 KM/Hr
6	Turn	Medium	11.38295, 77.90001	30 KM/Hr
7	Turn	High	11.38186, 77.90258	15 KM/Hr
8	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
9	Turn	Medium	11.36745, 77.93951	30 KM/Hr
10	Turn	Medium	11.41354, 78.08075	30 KM/Hr
11	Turn	High	11.41447, 78.08063	15 KM/Hr
12	Turn	High	11.41475, 78.08087	15 KM/Hr
13	Turn	High	11.42964, 78.11587	15 KM/Hr
14	Turn	Medium	11.42447, 78.15264	30 KM/Hr
15	Turn	Medium	11.43142, 78.18808	30 KM/Hr
16	Turn	Medium	11.43426, 78.20571	30 KM/Hr
17	Turn	High	11.43542, 78.20754	15 KM/Hr
18	Blind Spot	Blind Spot	11.45100, 78.20385	10 KM/Hr
19	Turn	Medium	11.44993, 78.22152	30 KM/Hr
20	Turn	Medium	11.44951, 78.22263	30 KM/Hr
21	Turn	High	11.44797, 78.22314	15 KM/Hr
22	Turn	Medium	11.44756, 78.22414	30 KM/Hr
23	Turn	Medium	11.45353, 78.23584	30 KM/Hr
24	Turn	Medium	11.49835, 78.32417	30 KM/Hr
25	Turn	Medium	11.50506, 78.33087	30 KM/Hr
26	Turn	Medium	11.50865, 78.33966	30 KM/Hr
27	Turn	High	11.51183, 78.34416	15 KM/Hr
28	Turn	Medium	11.51042, 78.34865	30 KM/Hr
29	Turn	Medium	11.50973, 78.34962	30 KM/Hr
30	Turn	Medium	11.50909, 78.35278	30 KM/Hr
31	Turn	Medium	11.48668, 78.36892	30 KM/Hr
32	Turn	Medium	11.47991, 78.38028	30 KM/Hr
33	Turn	Medium	11.46964, 78.39358	30 KM/Hr
34	Turn	Medium	11.46790, 78.39420	30 KM/Hr
35	Turn	Medium	11.46691, 78.39407	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
36	Turn	Medium	11.46466, 78.39289	30 KM/Hr
37	Turn	Medium	11.46410, 78.39311	30 KM/Hr
38	Turn	Medium	11.45677, 78.40081	30 KM/Hr
39	Turn	High	11.45597, 78.40096	15 KM/Hr
40	Turn	Medium	11.45563, 78.40251	30 KM/Hr
41	Turn	Medium	11.45658, 78.40441	30 KM/Hr
42	Turn	Medium	11.46740, 78.45082	30 KM/Hr
43	Turn	Medium	11.46728, 78.45442	30 KM/Hr
44	Turn	Medium	11.46798, 78.45523	30 KM/Hr
45	Blind Spot	Blind Spot	11.46798, 78.45844	10 KM/Hr
46	Blind Spot	Blind Spot	11.46857, 78.45844	10 KM/Hr
47	Turn	Medium	11.46835, 78.45959	30 KM/Hr
48	Turn	Medium	11.46800, 78.45987	30 KM/Hr
49	Turn	Medium	11.46461, 78.47397	30 KM/Hr
50	Turn	Medium	11.46419, 78.47445	30 KM/Hr
51	Turn	Medium	11.46211, 78.48159	30 KM/Hr
52	Turn	Medium	11.46184, 78.48270	30 KM/Hr
53	Turn	High	11.46106, 78.48357	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
10	hospital	Government Hospital, Namagiripettai	11.4569484, 78.265263	30 km/h	Medium
11	clinic	Shreetejo Ayurveda Clinic	11.444915, 78.485868	30 km/h	Medium
12	hospital	The Chirist Hospital	11.4443163, 78.4862134	30 km/h	Medium
13	hospital	Annai Hospital	11.4437321, 78.4858654	30 km/h	Medium
14	hospital	Vinola Hospital	11.4439899, 78.4857585	30 km/h	Medium
15	clinic	Sabeetha Surgical Clinic	11.4421254, 78.4867012	30 km/h	Medium
16	hospital	Dr. Ramu Life Care Hospital	11.4426079, 78.4879762	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	marketplace	Farmer Market	11.4146849, 78.0805463	30 km/h	Medium
7	school	Government Higher Secondary School	11.4150045, 78.0838172	30 km/h	Medium
8	school	Government school	11.4248027, 78.1323902	30 km/h	Medium
9	college	Paavai Arts And Science College For Women	11.4262443, 78.1546987	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44019, 77.87547



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39685, 77.88861



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.38376, 77.89914



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38342, 77.89975



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38295, 77.90001



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38186, 77.90258



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.37841, 77.90140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.36745, 77.93951



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.41354, 78.08075



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41447, 78.08063



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41475, 78.08087



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.42964, 78.11587



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.42447, 78.15264



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43142, 78.18808



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.43542, 78.20754



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45100, 78.20385



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44993, 78.22152



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44951, 78.22263



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44797, 78.22314



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44756, 78.22414



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45353, 78.23584



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49835, 78.32417



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50506, 78.33087



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50865, 78.33966



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.51183, 78.34416



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51042, 78.34865



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50973, 78.34962



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50909, 78.35278



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.47991, 78.38028



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46964, 78.39358



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46790, 78.39420



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46691, 78.39407



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46466, 78.39289



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46410, 78.39311



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45597, 78.40096



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45563, 78.40251



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45658, 78.40441



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46740, 78.45082



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46728, 78.45442



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46798, 78.45523



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.46798, 78.45844



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.46857, 78.45844



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46835, 78.45959



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46800, 78.45987



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46461, 78.47397



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46419, 78.47445



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46211, 78.48159



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46184, 78.48270



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46106, 78.48357
